Magnetic dipole moment

$$\vec{A}(\vec{R}) = \frac{\mu_0}{4\pi} \int d\vec{r} \, \frac{\vec{j}(\vec{r})}{|\vec{R}-\vec{r}|}$$

let $R \gg a$ ← size of the region where charges move

Then we can expand ~~the dbbb~~ in powers $\frac{a}{R}$.

$$\frac{1}{|\vec{R}-\vec{r}|} = \frac{1}{R} + r_i \frac{\partial}{\partial R_i} \frac{1}{|\vec{R}-\vec{r}|} \Big|_{r=0} + \ldots$$

$$= \frac{1}{R} - r_i \underbrace{\frac{\partial}{\partial R_i} \frac{1}{R}}_{-\frac{R_i}{R^3}} = \frac{1}{R} + \frac{r_i R_i}{R^3} = \frac{1}{R} + \frac{\vec{r}\cdot\hat{n}}{R^2}$$

$$\hat{n} = \frac{\vec{R}}{R}$$

$$\vec{A}(\vec{R}) \simeq \frac{\mu_0}{4\pi R} \underbrace{\int dV \, \vec{j}(\vec{r})}_{\substack{\emptyset \\ \text{current} \\ \text{over closed} \\ \text{loops}}} + \frac{\mu_0}{4\pi R^2} \int dV \, \vec{j} \, (\hat{n}\cdot\vec{r})$$

OR, Point charges point of view:

$$\vec{A}(\vec{R}) = \frac{\mu_0}{4\pi} \sum_n \frac{q_n \vec{v}_n}{|\vec{R}-\vec{r}_n|}$$

Then the leading term of the expansion is just

$$\vec{A}(\vec{R}) \simeq \frac{\mu_0}{4\pi R} \underbrace{\sum_n q_n \vec{v}_n}_{\frac{d}{dt}\sum_n q_n \vec{r}_n = \dot{\vec{d}}} + \frac{\mu_0}{4\pi R^2} \sum_n q_n \vec{v}_n (\vec{n}\cdot\vec{r}_n)$$

time-average of that over the period is zero.

full time derivative $\to 0$

$$\sum q\,\vec{v}(\hat{n}\cdot\vec{r}) = \frac{1}{2}\frac{d}{dt}\sum q_n \vec{r}_n(\hat{n}\cdot\vec{r}_n) - \frac{1}{2}\sum q_n \vec{r}_n(\hat{n}\cdot\vec{v}_n)$$

$$+\frac{1}{2}\sum q_n \vec{v}_n(\hat{n}\cdot\hat{r}_n) = \frac{1}{2}\sum q\left\{\vec{v}(\hat{n}\cdot\vec{r}) - \vec{r}(\hat{n}\cdot\vec{v})\right\}$$

$$A\times(B\times C) = B(A\cdot C) - C(A\cdot B)$$

$$= \frac{1}{2}\sum q\,(\vec{r}\times\vec{v})\times\hat{n} = \frac{1}{2}\sum q\,\hat{n}\times(\vec{v}\times\vec{r})$$

$$\hat{n}\times(\vec{v}\times\vec{r}) = \vec{v}(\hat{n}\cdot\vec{r}) - \vec{r}(\hat{n}\cdot\vec{v})$$

$$\Rightarrow \vec{A}(\vec{R}) = \frac{\mu_0}{4\pi}\frac{\vec{m}\times\hat{n}}{R^2} = \frac{\mu_0}{4\pi}\frac{\vec{m}\times\vec{R}}{R^3}$$

time-averaged

mag. moment $\quad \vec{m} = \frac{1}{2}\sum q\,\vec{r}\times\vec{v} = \frac{1}{2}\int d\vec{r}\;\vec{r}\times\vec{j}$ axial vector

Check current loop: $\vec{m} = \frac{1}{2}\int \underbrace{dV}_{da\,dl}\;\vec{r}\times\vec{j} =$

$$\vec{j}\,dV \to I\,d\vec{l}$$

$$\vec{m} = \frac{1}{2}\int \vec{r}\times\vec{j}\,dV = \frac{1}{2}I\int \vec{r}\times d\vec{l} = I\int d\vec{a}$$

Stokes $\displaystyle\int_S (\vec{\nabla}\times\vec{F})\cdot d\vec{a} = \oint_\Gamma \vec{F}\cdot d\vec{l}$ $\qquad d\vec{l} = \hat{x}\,dx + \hat{y}\,dy + \hat{z}\,dz$

let $\vec{F} = \frac{1}{2}(-y, x, 0) \Rightarrow \vec{\nabla}\times\vec{F} = \frac{1}{2}\begin{vmatrix}\hat{x} & \hat{y} & \hat{z}\\ \partial_x & \partial_y & \partial_z\\ -y & x & 0\end{vmatrix} = \hat{z}\left(\frac{1}{2} - \left(-\frac{1}{2}\right)\right) = \hat{z}$

$$\Rightarrow \int \hat{z}\cdot d\vec{a} = \frac{1}{2}\int(-y\hat{x} + x\hat{y})\cdot d\vec{l} = \frac{1}{2}\int(x\,dy - y\,dx)$$

Consider $m_z = \frac{1}{2} I \int \left(\vec{r} \times d\vec{l}\right)_z = I \int \left(\frac{1}{2} x\,dy - \frac{1}{2} y\,dx\right)$

$$= I \oint \vec{F} \cdot d\vec{l}$$

where $\vec{F} = \frac{1}{2}(-y, x, 0)$.

$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ x & y & z \\ dx & dy & dz \end{vmatrix}_z = \hat{z}\,(x\,dy - y\,dx)$$

$$\Rightarrow \quad m_z = I \int da_z \quad \Rightarrow \quad \vec{m} = I \int d\vec{a}.$$

<u>Mag. field</u> $\quad \vec{B} = \vec{\nabla} \times \vec{A}$

$$B_i = \varepsilon_{ijk} \frac{\partial}{\partial x_j} A_k = \frac{\mu_0}{4\pi} \varepsilon_{ijk} \frac{\partial}{\partial x_j} \left(\quad \right)_k$$

$$= \frac{\mu_0}{4\pi} \varepsilon_{ijk} \frac{\partial}{\partial x_j} \left(\varepsilon_{klm} \frac{m_l x_m}{R^3} \right) =$$

$$= \frac{\mu_0}{4\pi} \left(\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl} \right) m_l \left(\frac{\delta_{jm}}{R^3} - x_m \frac{3}{R^4} \underbrace{\frac{\partial R}{\partial x_j}}_{\frac{x_j}{R}} \right)$$

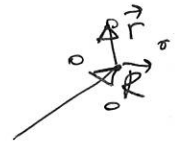
$$= \frac{\mu_0}{4\pi} \left(\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl} \right) m_l \frac{\delta_{jm} - 3 n_j n_m}{R^3}$$

$$\vec{B} = \frac{\mu_0}{4\pi} \frac{3(\vec{m} \cdot \hat{n})\,\hat{n} - \vec{m}}{R^3}$$

Magnetic dipole in external field (very important physics problem, relevant to ESR, NMR ...)

$$\vec{F} = \sum q\, \vec{v} \times \vec{B}(\vec{R}+\vec{r})$$

charges



$$\frac{\partial}{\partial r_j} = \frac{\partial}{\partial R_j}$$

$$\vec{B}(\vec{R}+\vec{r}) \cong \vec{B}(\vec{R}) + r_j \frac{\partial}{\partial r_j} \vec{B}(\vec{R}) = \vec{B}(\vec{R}) + (\vec{r}\cdot\vec{\nabla}_R)\vec{B}(\vec{R})$$

$$\vec{F} = \sum \underbrace{q\,\vec{v} \times \vec{B}(\vec{R})}_{\dot{\vec{d}} \times \vec{B}(\vec{R}) \to 0} + \sum q\,\vec{v} \times (\vec{r}\cdot\vec{\nabla}_R)\vec{B}(\vec{R})$$

when we average over the period

$$F_i = \sum q\left(\vec{v} \times r_\ell \frac{\partial}{\partial R_\ell} \vec{B}(\vec{R})\right)_i =$$

$$= \varepsilon_{ijk} \sum q\, v_j\, r_\ell \frac{\partial B_k}{\partial R_\ell} = \varepsilon_{ijk} \frac{\partial B_k}{\partial R_\ell} \sum q\, v_j\, r_\ell$$

$$B_k = (\vec{\nabla} \times \vec{A})_k = \varepsilon_{knm} \frac{\partial A_m}{\partial R_n}$$

$$\Rightarrow \varepsilon_{ijk} \frac{\partial B_k}{\partial R_\ell} = \underbrace{\varepsilon_{ijk}\, \varepsilon_{knm}}_{\varepsilon_{kij}\, \varepsilon_{knm} = \delta_{in}\delta_{jm} - \delta_{im}\delta_{jn}} \frac{\partial^2 A_m}{\partial R_\ell \partial R_n} = \frac{\partial^2 A_j}{\partial R_\ell \partial R_i} - \frac{\partial^2 A_i}{\partial R_\ell \partial R_j}$$

$$\Rightarrow F_i = \left(\frac{\partial^2 A_j}{\partial R_\ell \partial R_i} - \frac{\partial^2 A_i}{\partial R_\ell \partial R_j}\right) \sum q\, v_j\, r_\ell$$

Next we notice that $\sum q\, v_j r_\ell = -\sum q\, v_\ell r_j$

This is because $\sum q\left(v_j r_\ell + v_\ell r_j\right) = \frac{d}{dt}\sum q\, r_i r_\ell$

and total t-derivative vanishes after averaging over the period of motion. $\Rightarrow \sum q\, v_j r_\ell$ is anti-symmetric.

But
$$\underbrace{\frac{\partial^2 A_i}{\partial R_\ell \partial R_j}}_{\substack{\text{symmetric}\\ \text{in } j,\ell}} \underbrace{\sum q\, v_j r_\ell}_{\text{antisymmetric in } j,\ell} = 0.$$

$$\Rightarrow F_i = \frac{\partial^2 A_j}{\partial R_\ell \partial R_i}\sum q\, v_j r_\ell = \frac{\partial}{\partial R_i}\left(\frac{\partial A_j}{\partial R_\ell}\sum q\, v_j r_\ell \cdot \frac{1}{2}\right.$$
$$\left. + \frac{\partial A_\ell}{\partial R_j}\underbrace{\sum q\, v_\ell r_j}_{-\sum q\, v_j r_\ell}\cdot\frac{1}{2}\right) = \frac{\partial}{\partial R_i}\left(\frac{\partial A_j}{\partial R_\ell} - \frac{\partial A_\ell}{\partial R_j}\right)\frac{1}{2}\sum q\, v_j r_\ell$$

But
$$-\left(\frac{\partial A_j}{\partial R_\ell} - \frac{\partial A_\ell}{\partial R_j}\right) = \varepsilon_{kj\ell} B_k = \varepsilon_{kj\ell}\left(\varepsilon_{knm}\frac{\partial A_m}{\partial R_n}\right) =$$
$$= \left(\delta_{jn}\delta_{\ell m} - \delta_{jm}\delta_{\ell n}\right)\frac{\partial A_m}{\partial R_n} = \frac{\partial A_\ell}{\partial R_j} - \frac{\partial A_j}{\partial R_\ell}$$

$$F_i = \frac{\partial}{\partial R_i}\left(-\varepsilon_{kj\ell} B_k\right)\frac{1}{2}\sum q\, v_j r_\ell = \frac{\partial}{\partial R_i}\left(B_k \mu_k\right)$$

where
$$\mu_k = \frac{1}{2}\sum\left(q\,\vec{r}\times\vec{v}\right)_k = \frac{1}{2}\sum q\, \varepsilon_{k\mu\nu} r_\mu v_\nu =$$
$$= -\frac{1}{2}\sum q\, v_\nu r_\mu\, \varepsilon_{k\nu\mu}$$

$\Rightarrow \vec{F} = \vec{\nabla}\left(\vec{\mu}\cdot\vec{B}\right)$ force appears only in inhomogeneous magnetic field.

Therefore $\mathcal{E} = -\vec{\mu} \cdot \vec{B}$.

Quite similar to $\mathcal{E} = -\vec{d} \cdot \vec{E}$ for electric dipole.

For 2 magnetic dipoles: $\mathcal{E} = -\vec{\mu}_1 \cdot \vec{B}_2 = \frac{\mu_0}{4\pi} \frac{\vec{\mu}_1 \cdot \vec{\mu}_2 - 3(\vec{\mu}_1 \cdot \hat{n})(\vec{\mu}_2 \cdot \hat{n})}{R^3}$.

Magnetic dipole in a field. Bloch equations.

Uniform field $\vec{B}$: $\mathcal{E} = -\vec{\mu}\cdot\vec{B}$, $\vec{F} = \vec{\nabla}(\vec{\mu}\cdot\vec{B})$

$\Rightarrow$ magnetic dipole experiences force only if $\vec{B}$ is non-uniform.

However there is <u>torque</u> $\vec{\tau} = \vec{r}\times\vec{f}$.

$$\Rightarrow \quad \tau = \underbrace{\sum q\vec{r}\times\vec{E}}_{\text{El. field}} + \underbrace{\sum q\vec{r}\times(\vec{v}\times\vec{B})}_{\text{Lorentz force}}$$

Note $\sum q\vec{r} = \vec{d}$ = electric dipole moment

$$\Rightarrow \vec{\tau}_E = \vec{d}\times\vec{E}.$$

Look on $\vec{\tau}_B = \sum q\,\vec{r}\times(\vec{v}\times\vec{B}) = \sum q\,\vec{v}\,(\vec{r}\cdot\vec{B}) - \vec{B}\sum q\,\vec{v}\cdot\vec{r}$

But $\sum q\,\vec{v}\cdot\vec{r} = \frac{d}{dt}\frac{1}{2}\sum q r^2 \Rightarrow$ averages to zero for spatiotemporal currents.

Next $\sum q\,\vec{v}(\vec{r}\cdot\vec{B}) = \hat{e}_j \sum q\, v_j\, r_i B_i$

But $v_j r_i B_i = \frac{1}{2} v_j r_i B_i + \frac{1}{2}(v_j r_i B_i + v_i r_j B_i) - \frac{1}{2} v_i r_j B_i$

$$= \frac{1}{2}(v_j r_i B_i - v_i r_j B_i) + \frac{1}{2} B_i \underbrace{\frac{d}{dt}(r_j r_i)}_{\text{zero after averaging over the period.}}$$

$$\Rightarrow \sum q\,\vec{v}(\vec{r}\cdot\vec{B}) = \frac{1}{2}\sum q\,\vec{v}(\vec{r}\cdot\vec{B}) - q\vec{r}(\vec{v}\cdot\vec{B}) =$$

$$= \frac{1}{2}\sum q(\vec{r}\times\vec{v})\times\vec{B} = -\frac{1}{2}\sum q\,\vec{B}\times(\vec{r}\times\vec{v})$$

$$= -\frac{1}{2}\sum\left(q\,\vec{r}(\vec{B}\cdot\vec{v}) - q\,\vec{v}(\vec{B}\cdot\vec{r})\right).$$

$$\Rightarrow \vec{\tau}_B = \frac{1}{2}\sum q(\vec{r}\times\vec{v})\times\vec{B} = \vec{\mu}\times\vec{B}$$

$\Rightarrow$ for stationary system of currents $\vec{\tau}_E = 0$,

$$\vec{\tau}_B = \vec{\mu}\times\vec{B}$$

But we know that $\vec{\tau} = \frac{d\vec{L}}{dt} = \frac{d}{dt}\sum \vec{r}\times\vec{p}$ angular momentum

$$\Rightarrow \frac{d\vec{L}}{dt} = \vec{\mu}\times\vec{B}$$

$$\begin{cases} \vec{L} = \sum \vec{r}\times m\vec{v} = \sum m\,\vec{r}\times\vec{v} \\ \vec{\mu} = \frac{1}{2}\sum q\,\vec{r}\times\vec{v} \end{cases}$$

$\Rightarrow$ in a system of particles with the same charge-to-mass $\frac{q}{m}$

$$\vec{\mu} = \frac{1}{2}\frac{q}{m}\sum m\vec{r}\times\vec{v} = \frac{q}{2m}\vec{L} = \gamma\vec{L}$$

$$\Rightarrow \quad \gamma\frac{d\vec{L}}{dt} = \gamma\vec{\mu}\times\vec{B}$$

$$\boxed{\frac{d\vec{\mu}}{dt} = \gamma\vec{\mu}\times\vec{B}} \qquad \gamma = \frac{q}{2m} = \text{gyromagnetic ratio.}$$

$\vec{\mu}$ precesses with angular frequency $\vec{\omega}_0 = -\gamma\vec{B}$ = Larmor frequency.

$$\omega_0 = |\vec{\omega}_0| = \gamma B = \frac{qB}{2m}$$

(Note: cyclotron freq. is different $\omega_c = \frac{qB}{m} = 2\omega_0$)

<u>Cyclotron</u> motion: charge q rotates in XY-plane around z-axis ($\vec{B} = B\hat{z}$) due to Lorentz force $q\vec{v}\times\vec{B}$.

<u>Larmor</u> (precessional) motion:

rotation is intrinsic (it is the motion that creates $\vec{\mu}$ and $\vec{L}$)

and $\vec{B}$ simply induces precession of $\vec{\mu}$.

(Electrons: $q = -e \Rightarrow \gamma = -\frac{e}{2m} < 0$.)

Quantum mechanics: total ang. momentum $= \vec{L} + \vec{S}$ (orbital, spin)

$\Rightarrow \gamma = -\frac{ge}{2m}$ (g-factor $= 2$)

$$\frac{d}{dt}\vec{\mu} = \gamma \vec{\mu} \times \vec{B} \Rightarrow \vec{B} \cdot \frac{d\vec{\mu}}{dt} = 0 \Rightarrow \underline{\vec{\mu} \cdot \vec{B} = \text{const}}$$

conservation of energy!

This says that angle θ between $\vec{\mu}$ and $\vec{B}$ is constant because $\mathcal{E} = -\vec{\mu}\cdot\vec{B} = -\mu B \cos\theta$

energy is conserved.

$\vec{\mu}$ in solids:

But of course in reality energy from $\vec{\mu}$ gets transferred, albeit slowly, to other degrees of freedom.

So that $\mathcal{E}$ is not strictly conserved. This allows $\vec{\mu}$ to slowly <u>relax</u> towards $\vec{B}$ so that at the end of the day

$$\mathcal{E} = -\mu B < -\mu B \cos\theta \text{ for } \theta \neq 0.$$

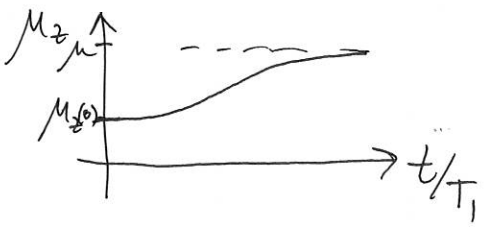
This introduces longitudinal relaxation time T_1

$$(1) \quad \frac{d\mu_z(t)}{dt} = \frac{\mu - \mu_z(t)}{T_1} \Rightarrow \mu_z(t) = \underset{\substack{\uparrow \\ \text{total} \\ \mu = |\vec{\mu}|.}}{\mu}\left(1 - e^{-t/T_1}\right) + \underset{\substack{\uparrow \\ \mu_z(t=0)}}{\mu(0)}\, e^{-t/T_1}$$

Since $\mu = \vec{\mu}_\perp^2 + \mu_z^2$

↳ component of $\vec{\mu}$ perpendicular to $\vec{B}$.

As $\mu_z(t)$ increases from $\mu_z(t=0)$ towards $\mu_z(t=\infty) = \mu$



the other two components

$$\mu_\perp = |\vec{\mu}_\perp| = \sqrt{\mu^2 - \mu_z^2} \text{ approaches } 0.$$

Vector $\vec{\mu}_\perp$ rotates in xy-plane with constant frequency $\omega_0 = -\gamma B$.

(2) There also is <u>transverse relaxation</u> time T_2 $(T_2 \ll T_1)$

$$\frac{d\vec{\mu}_\perp}{dt} = \gamma \vec{\mu}_\perp \times \vec{B} - \frac{\vec{\mu}_\perp}{T_2}. \quad (1)+(2) = \text{Bloch equations}$$